

A GENERALISED RATIO AND PRODUCT TYPE ESTIMATOR FOR THE POPULATION MEAN IN STRATIFIED RANDOM SAMPLING

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Key words: Auxiliary variable; combined ratio estimator; efficiency; mean square error; ratio and product-type estimators; stratified random sampling.

Summary: For estimating the population mean, a generalised ratio and product type estimator has been proposed by using known information in stratified random sampling. The expressions for mean square error (MSE) and its minimum values have been obtained. The comparison of the proposed estimator with the estimator defined by Kadilar and Cingi (2005) have been made theoretically. The results have also been illustrated numerically in some special cases. It has been shown that the proposed estimator is always better than the combined ratio estimator and the Kadilar and Cingi (2005) estimator.

1. Introduction

In sampling theory, information on an auxiliary variable has been used at the design stage of the survey, i.e. in stratifying the population, as well as at the sample selection stage, i.e. in selection of sample units by the use of unequal probability sampling based on the size of units with respect to auxiliary variable. It has also been used at the estimation stage, i.e. in the use of ordinary ratio, product, difference and regression estimators etc.

Using an auxiliary information at the estimation stage, Kadilar and Cingi (2003) have defined ratio estimators of the population mean in stratified random

AMS: 62D05

sampling similar to the estimators defined by Sisodia and Dwivedi (1981), Singh and Kakran (1993), and Upadhyaya and Singh (1999). They have shown that estimators defined by them are more efficient than the traditional combined ratio estimator under some conditions. In 1989, Prasad defined some improved ratio-type estimators of the population mean. Following Prasad (1989), Kadilar and Cingi (2005) defined a new ratio estimator of the population mean $\bar{Y}$ of the study variable y using stratified random sampling as

$$\bar{y}_{stp} = \kappa^* \bar{y}_{RC}. \quad (1.1)$$

Here $\bar{y}_{RC}$ is the usual combined ratio estimator of population mean $\bar{Y}$ in stratified random sampling and κ^* is any constant. They have shown that the estimator $\bar{y}_{stp}$ defined in (1.1) is more efficient than the usual combined ratio estimator $\bar{y}_{RC}$.

In the present paper, we propose a generalised ratio and product type estimator of population mean under stratified random sampling using known information on parameters of auxiliary variable x . The estimators defined by Kadilar and Cingi (2003) are particular cases of the proposed estimator. The expressions for mean square error and its minimum values have been obtained. The results have also been illustrated numerically as well as graphically.

2. Sampling design and terminology

Let a population of N units be divided into k strata and let simple random samples of size n_h be drawn from the h -th stratum of size N_h with $\sum_{h=1}^k N_h = N$. Let Y_{hi} and X_{hi} denote the value of the variable under study y , and the auxiliary variable x on the i -th unit of the population in the h -th stratum, respectively. The corresponding small letters denote the values

in the sample. Suppose the population mean $\bar{X}$ of auxiliary variable x is known and the combined ratio estimate of population mean ($\bar{Y}$) in stratified random sampling is defined as

$$\bar{y}_{RC} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X} = \hat{R}_c \bar{X} \quad (2.1)$$

where $\bar{y}_{st} = \sum_{h=1}^k w_h \bar{y}_h$, $\bar{x}_{st} = \sum_{h=1}^k w_h \bar{x}_h$, $\bar{X} = \sum_{h=1}^k w_h \bar{X}_h$ and $\bar{Y} = \sum_{h=1}^k w_h \bar{Y}_h$ with $w_h = \frac{N_h}{N}$ the stratum weight.

Here $\bar{y}_h, \bar{x}_h$ and $\bar{Y}_h, \bar{X}_h$ denote the sample means and population means of variables y, x respectively in the h -th stratum.

Up to first-order of approximation, the MSE of the combined ratio estimator is

$$MSE(\bar{y}_{RC}) = \sum_{h=1}^k w_h^2 \gamma_h (S_{y_h}^2 - 2R S_{y_h x_h} + R^2 S_{x_h}^2) \quad (2.2)$$

Here $\gamma_h = \frac{1 - (n_h/N_h)}{n_h}$ and $R = \frac{\bar{Y}}{\bar{X}}$; $S_{y_h}^2$ and $S_{x_h}^2$ denote the population variances of the variables y and x respectively in the h -th stratum and $S_{y_h x_h}$ is the population covariance between them.

3. The proposed estimator

In stratified random sampling, when information on population parameters θ_h and ϕ_h along with population mean $\bar{X}_h$ of the auxiliary variable x are known in advance, we propose a generalised ratio and product-type estimator of the population mean

$$\hat{\bar{y}}_{jw} = \bar{y}_{st} \left(\frac{\bar{x}'_{st}}{\bar{X}'_{st}} \right)^\alpha \quad (3.1)$$

Here $\bar{x}'_{st} = \sum_{h=1}^k w_h (\theta_h \bar{x}_h + \phi_h)$, $\bar{X}'_{st} = \sum_{h=1}^k w_h (\theta_h \bar{X}_h + \phi_h)$, $\bar{y}_{st} = \sum_{h=1}^k w_h \bar{y}_h$, α is any arbitrary constant, and θ_h and ϕ_h are any known parameters of the auxiliary variable.

Kadilar and Cingi (2003) have defined the following ratio-type estimators by using the known information on the coefficient of variation (C_{xh}), the population mean $\bar{X}_h$ and the coefficient of kurtosis ($\beta_{2h}(x)$) of the auxiliary variable x in the h -th stratum as

$$\begin{aligned}
 \text{(i)} \quad \bar{y}_{stSD} &= \bar{y}_{st} \frac{\sum_{h=1}^k w_h (\bar{X}_h + C_{xh})}{\sum_{h=1}^k w_h (\bar{x}_h + C_{xh})} \\
 \text{(ii)} \quad \bar{y}_{stSK} &= \bar{y}_{st} \frac{\sum_{h=1}^k w_h (\bar{X}_h + \beta_{2h}(x))}{\sum_{h=1}^k w_h (\bar{x}_h + \beta_{2h}(x))} \\
 \text{(iii)} \quad \bar{y}_{stUS1} &= \bar{y}_{st} \frac{\sum_{h=1}^k w_h (\bar{X}_h \beta_{2h}(x) + C_{xh})}{\sum_{h=1}^k w_h (\bar{x}_h \beta_{2h}(x) + C_{xh})} \\
 \text{(iv)} \quad \bar{y}_{stUS2} &= \bar{y}_{st} \frac{\sum_{h=1}^k w_h (\bar{X}_h C_{xh} + \beta_{2h}(x))}{\sum_{h=1}^k w_h (\bar{x}_h C_{xh} + \beta_{2h}(x))}
 \end{aligned}$$

The estimators $\bar{y}_{stSD}$, $\bar{y}_{stSK}$, $\bar{y}_{stUS1}$ and $\bar{y}_{stUS2}$ defined by Kadilar and Cingi (2003) can be obtained from the proposed estimator $\hat{\bar{y}}_{jw}$ defined in (3.1) by taking

- (i) $\alpha = 1$, $\theta_h = 1$, $\phi_h = C_{xh}$
- (ii) $\alpha = 1$, $\theta_h = 1$, $\phi_h = \beta_{2h}(x)$
- (iii) $\alpha = 1$, $\theta_h = \beta_{2h}(x)$, $\phi_h = C_{xh}$ and

(iv) $\alpha = 1$, $\theta_h = C_{xh}$, $\phi_h = \beta_{2h}(x)$ respectively.

To obtain the bias and mean square error (MSE) of the proposed estimator $\hat{y}_{jw}$, we first define

$$\varepsilon = \frac{\bar{y}_{st}}{\bar{Y}} - 1 \quad \text{and} \quad \eta = \frac{\bar{x}'_{st}}{\bar{X}'_{st}} - 1 \quad (3.2)$$

such that $E(\varepsilon) = E(\eta) = 0$.

Expanding the right-hand side of (3.1) in terms of ε and η , we get

$$\hat{y}_{jw} = \bar{Y} \left[1 + \varepsilon + \alpha\eta + \alpha\varepsilon\eta + \frac{\alpha(\alpha-1)}{2}\eta^2 + \dots \right] \quad (3.3)$$

Taking expectation on both sides of (3.3) and retaining terms up to first-order of approximation, we have

$$\begin{aligned} E(\hat{y}_{jw}) &= \bar{Y} + \bar{Y} \left[\frac{\alpha}{\bar{Y}\bar{X}'_{st}} \sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right. \\ &\quad \left. + \frac{\alpha(\alpha-1)}{2} \frac{1}{\bar{X}'_{st}{}^2} \sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2 \right] \\ \Rightarrow Bias(\hat{y}_{jw}) &= \bar{Y} \left[\frac{\alpha}{\bar{Y}\bar{X}'_{st}} \sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right. \\ &\quad \left. + \frac{\alpha(\alpha-1)}{2} \frac{1}{\bar{X}'_{st}{}^2} \sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2 \right] \end{aligned}$$

Up to first-order of approximation, the MSE of proposed estimator $\hat{y}_{jw}$ is

$$\begin{aligned}
MSE(\hat{\bar{y}}_{jw}) &= \sum_{h=1}^k \gamma_h w_h^2 \\
&\quad \cdot \left[\left(S_{y_h}^2 + \left(\frac{\bar{Y}}{\bar{X}_{st}'} \right)^2 \alpha^2 \theta_h^2 S_{x_h}^2 + 2\alpha \left(\frac{\bar{Y}}{\bar{X}_{st}'} \right) \theta_h S_{y_h x_h} \right) \right] \\
&= \sum_{h=1}^k \gamma_h w_h^2 (S_{y_h}^2 + R_{jw}^2 \alpha^2 \theta_h^2 S_{x_h}^2 + 2\alpha R_{jw} \theta_h S_{y_h x_h}) \quad (3.4)
\end{aligned}$$

where

$$R_{jw}^2 = \frac{\bar{Y}}{\bar{X}_{st}'} = \frac{\bar{Y}}{\sum_{h=1}^k w_h (\theta_h \bar{X}_h + \phi_h)}. \quad (3.5)$$

Differentiating expression (3.4) with respect to α and equating to zero, we get

$$(\alpha)_{opt} = - \frac{1}{R_{jw}} \frac{\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h}}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2}. \quad (3.6)$$

Substituting the optimum value of α obtained in (3.6) in (3.4), the MSE of $\hat{\bar{y}}_{jw}$ is

$$MinMSE(\hat{\bar{y}}_{jw}) = \sum_{h=1}^k \gamma_h w_h^2 S_{y_h}^2 - \frac{\left(\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right)^2}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2} \quad (3.7)$$

Special case: For $\theta_h = \frac{S_{y_h x_h}}{S_{x_h}^2} = \beta_{y_h x_h}$, the $MinMSE(\hat{\bar{y}}_{jw})$ takes the form

$$MinMSE(\hat{\bar{y}}_{jw}) = \sum_{h=1}^k \gamma_h w_h^2 S_{y_h}^2 (1 - \rho_h^2) \quad (3.8)$$

where ρ_h is the correlation coefficient between y and x in the h -th stratum.

The results are summarised in the following theorem:

Theorem Up to first-order of approximation, the MSE of $\hat{\bar{y}}_{jw}$ is

$$MSE(\hat{\bar{y}}_{jw}) = \sum_{h=1}^k \gamma_h w_h^2 (S_{y_h}^2 + R_{jw}^2 \alpha^2 \theta_h^2 S_{x_h}^2 + 2\alpha R_{jw} \theta_h S_{y_h x_h})$$

which is minimised for

$$(\alpha)_{opt} = \frac{1}{R_{jw}} \frac{\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h}}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2}$$

and its minimum value is

$$MinMSE(\hat{\bar{y}}_{jw}) = \sum_{h=1}^k \gamma_h w_h^2 S_{y_h}^2 - \frac{\left(\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right)^2}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2}$$

4. Comparison

We compare the proposed estimator with the estimator given by Kadilar and Cingi (2005).

Up to first-order of approximation, the MSE of the optimum estimator of $\bar{y}_{stp}$ defined by Kadilar and Cingi (2005) is

$$MinMSE(\bar{y}_{stp}) = \sum_{h=1}^k \gamma_h w_h^2 (S_{y_h}^2 - 2R S_{y_h x_h} + R^2 S_{x_h}^2) \quad (4.1)$$

which shows that the optimum estimator $\bar{y}_{stp}$ is not more efficient than the combined ratio estimator $\bar{y}_{RC}$.

Using the expression (3.7) and (4.1), we obtain

$$\begin{aligned}
 MinMSE(\bar{y}_{stp}) - MinMSE(\hat{\bar{y}}_{jw}) &= \sum_{h=1}^k \gamma_h w_h^2 (S_{y_h}^2 - 2RS_{y_h x_h} + R^2 S_{x_h}^2) \\
 &\quad - \sum_{h=1}^k \gamma_h w_h^2 S_{y_h}^2 + \frac{\left(\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right)^2}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2} \\
 &= \sum_{h=1}^k \gamma_h w_h^2 (R^2 S_{x_h}^2 - 2RS_{y_h x_h}) \\
 &\quad + \frac{\left(\sum_{h=1}^k \gamma_h w_h^2 \theta_h S_{y_h x_h} \right)^2}{\sum_{h=1}^k \gamma_h w_h^2 \theta_h^2 S_{x_h}^2} \\
 &> 0 \quad \text{if} \quad \beta_{y_h x_h} < \frac{R}{2}
 \end{aligned}$$

Special case I: For $\theta_h = 1$, the expression (4.2) reduces to

$$\begin{aligned}
 MinMSE(\bar{y}_{stp}) - MinMSE(\hat{\bar{y}}_{jw}) &= \left(R \sum_{h=1}^k \gamma_h w_h^2 S_{x_h}^2 - \sum_{h=1}^k \gamma_h w_h^2 S_{x_h y_h} \right)^2 \\
 &\geq 0
 \end{aligned} \tag{4.3}$$

Special case II: For $\theta_h = \frac{S_{y_h x_h}}{S_{x_h}^2}$ the expression (4.2) reduces to

$$\begin{aligned}
 MinMSE(\bar{y}_{stp}) - MinMSE(\hat{\bar{y}}_{jw}) &= \sum_{h=1}^k \gamma_h w_h^2 \left(RS_{x_h} - \frac{S_{y_h x_h}}{S_{x_h}} \right)^2 \\
 &\geq 0
 \end{aligned} \tag{4.4}$$

In both the special cases, the expressions on the right-hand side in (4.3) and (4.4) are always positive. Hence we may conclude that the proposed estimator $\hat{\bar{y}}_{jw}$ is always more efficient than the Kadilar and Cingi (2005) estimator $\bar{y}_{stp}$.

5. Numerical example

To illustrate the results numerically, data on apple production amount (as variable of interest) and number of apple trees (as auxiliary variable) in 854 villages of Turkey in 1999 (Source: Institute of Statistics, Republic of Turkey) used by Kadilar and Cingi (2005) have been taken. The data have been stratified by regions of Turkey and from each stratum a random sample (villages) is drawn. The sample sizes are chosen by using the Neyman allocation,

$$n_h = n \frac{N_h S_h}{\sum_{h=1}^k N_h S_h}$$

The values of the parameters obtained are given in table 1 and efficiencies of the estimators are given in table 2.

Table 1 Values of parameters for a considered population

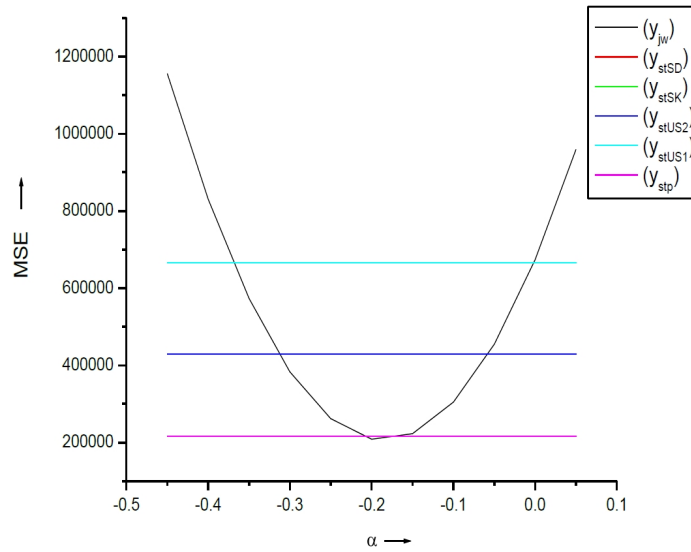
$N=854$	$N_1=106$	$N_2=106$	$N_3=94$	$N_4=171$	$N_5=204$	$N_6=173$
$n=140$	$n_1=9$	$n_2=17$	$n_3=38$	$n_4=67$	$n_5=7$	$n_6=2$
$\bar{X}=3760$	$\bar{X}_1=24375$	$\bar{X}_2=27421$	$\bar{X}_3=72409$	$\bar{X}_4=74365$	$\bar{X}_5=26441$	$\bar{X}_6=9844$
$\bar{Y}=2930$	$\bar{Y}_1=1536$	$\bar{Y}_2=2212$	$\bar{Y}_3=9384$	$\bar{Y}_4=5588$	$\bar{Y}_5=967$	$\bar{Y}_6=404$
$S_x=144794$	$S_{x_1}=49189$	$S_{x_2}=57461$	$S_{x_3}=160757$	$S_{x_4}=285603$	$S_{x_5}=45403$	$S_{x_6}=18794$
$S_y=17106$	$S_{y_1}=6425$	$S_{y_2}=11552$	$S_{y_3}=29907$	$S_{y_4}=28643$	$S_{y_5}=2390$	$S_{y_6}=946$
$C_x=3.85$	$C_{x_1}=2.02$	$C_{x_2}=2.10$	$C_{x_3}=2.22$	$C_{x_4}=3.84$	$C_{x_5}=1.72$	$C_{x_6}=1.91$
$\beta_x=312.07$	$\beta_{x_1}=25.71$	$\beta_{x_2}=34.57$	$\beta_{x_3}=26.14$	$\beta_{x_4}=97.60$	$\beta_{x_5}=27.47$	$\beta_{x_6}=28.10$
$R=0.07793$	$\rho_1=0.82$	$\rho_2=0.86$	$\rho_3=0.90$	$\rho_4=0.99$	$\rho_5=0.71$	$\rho_6=0.89$
$\kappa^*=0.975$	$\gamma_1=0.102$	$\gamma_2=0.049$	$\gamma_3=0.016$	$\gamma_4=0.009$	$\gamma_5=0.138$	$\gamma_6=0.006$
$\nu=215710.432$	$w_1^2=0.015$	$w_2^2=0.015$	$w_3^2=0.012$	$w_4^2=0.04$	$w_5^2=0.057$	$w_6^2=0.041$

Table 2 The efficiencies of estimators up to first-order of approximation

Estimator	MSE	Efficiency
$\bar{y}_{stUS2}$	665347.704	100
$\bar{y}_{stUS1}$	429527.205	154.90
$\bar{y}_{stSK}$	215841.602	308.25
$\bar{y}_{stSD}$	215717.639	308.43
$\bar{y}_{stp}$	215710.432	308.44
$\hat{\bar{y}}_{jw}$ (for $\theta_h = 1$)	202122.071	329.18
$\hat{\bar{y}}_{jw} \left(\text{for } \theta_h = \frac{S_{y_h x_h}}{S_{x_h}^2} \right)$	107065.782	621.43

For comparison purpose, results of the estimators have also been shown graphically.

Here $\theta_h = C_{xh}$ and $\phi_h = \beta_{2h}(x)$

**Figure 1.** Mean Square Errors (MSE) of different estimators with respect to the values of α

From the graph, we can find the range of variation of α in which the proposed estimator will be more efficient than the estimators proposed by Kadilar and Cingi (2003, 2005) under the given conditions.

Conclusion: From table 2, we see that the proposed estimator is almost six times more efficient than the estimator $\bar{y}_{stUS2}$ and almost three times more efficient than $\bar{y}_{stSK}$ and $\bar{y}_{stSD}$ defined by Kadilar and Cingi (2003) when θ_h is taken as the regression coefficient of y on x in the h -th stratum. It is also almost three times more efficient than the estimator $\bar{y}_{stp}$ defined by Kadilar and Cingi (2005). From the graph, we can also find the range of variation of α in which proposed estimator is more efficient than the estimators defined by Kadilar and Cingi (2003, 2005). Hence we conclude that the proposed estimator will be more efficient than the existing ones.

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